Safwat Ali Khan

@ Email in LinkedIn Github Website

Education

GEORGE MASON UNIVERSITY

PhD in Computer Science

Expected 2026

CGPA 3.88/4.00

UNIVERSITY OF DHAKA

BSc in Computer Science

2010-2014 Bangladesh

Graduate Courseworks

Software Testing • Algorithms • UI Design & Development • Deep Learning • Natural Language Processing

Skills

MACHINE LEARNING

Prompt Engineering • Fine-tuning LLMs • Deep Learning • Computer Vision • NLP • Information Retrieval • OpenCV • tensorflow • pyTorch • CLiP • Transformer

AUTOMATION TOOL:

UIAutomator • Selenium webdriver • JMeter • Google Monkey • APE

CORE EXPERTISE

GUI/UI Testing • Test Automation • Performance Testing • Automated Program Repair • Empirical Studies • Static & Dynamic Analysis

LANGUAGES

Python • Java • C/C++ • JavaScript

Research Talk

ICST DOCTORAL SYMPOSIUM

Apr, 2025 Napoli, Italy

Advancing Mobile UI Testing by Learning Screen Usage Symantics

FRAUNHOFER USA

Jan, 2023 Maryland, USA

Leveraging Machine Learning to bypass Android UI tarpits

Mentorship

ASSIP 2022, 2020

Co-mentor for Aspired Scientist Summer Internship Program

Service

JOURNAL REVIEWER

IEEE Trans. on Software Engineering (TSE) 2026

CO-REVIEWER

FSE 2024, ICSE 2023, SANER 2023, ASE 2022

STUDENT VOLUNTEER

ICST 2024

Research Publications

1. "Testing Practices, Challenges, and Developer Perspectives in Open-Source IoT Platforms" — Daniel Rodriguez-Cardenas, **Safwat Ali Khan**, Prianka Mandal, Adwait Nadkarni, Kevin Moran, Denys Poshyvanyk. In ICST 2025.
2. "AURORA: Navigating UI Tarpits via Automated Neural Screen Understanding" — **Safwat Ali Khan**, Wenyu Wang, Yiran Ren, Bin Zhu, Jiangfan Shi, Wing Lam, Kevin Moran. In ICST 2024.
3. "On Using GUI Interaction Data to Improve Text Retrieval-based Bug Localization" — Junayed Mahmud, Nadeeshan De Silva, **Safwat Ali Khan**, Seyed Hooman Mostafavi, S M Hasan Mansur, Oscar Chaparro, Andrian Marcus, Kevin Moran. In ICSE 2023.
4. "AVGUST: Automating Usage-based Test Generation from Videos of App Executions" — Yixue Zhao, Saghar Talebipour, Kesina Baral, Hyojae Park, Leon Yee, **Safwat Ali Khan**, Yuriy Brun, Nenad Medvidovic, Kevin Moran. In ESEC/FSE 2022.

Current Projects

- Building a **benchmark for automated program repair** of accessibility defects in software (manuscript **under review**).
- Conducting a **systematic literature review** of developer-facing accessibility tools, surveying guidelines, techniques, target populations, and evaluation methods across the SE lifecycle.

Fellowship

- Provost Scholarship — 2026
- Sustainability Summer Research Fellowship — 2024
- CS Research Initiation Award — 2020

Work Experience

GRADUATE ASSISTANT

George Mason University

Sep 2019 – Present Fairfax, VA, USA

Experience in academic research and teaching focusing software testing, data analysis, and program development.

- Conducting research on **automated mobile UI testing**, developing learning-based techniques to analyze large-scale execution traces and investigating limitations in smart home app testing.
- Advancing **software accessibility** research, including the automated program repair of accessibility defects and a systematic review of developer-facing accessibility tooling.
- Serving as a **Teaching Assistant**, mentoring students in Low-Level Programming (C, Java), Database Systems, and Data Structures courses.

INDUSTRY EXPERIENCE

Therap Services Ltd.

Jan 2018 – Jul 2019 Dhaka, Bangladesh

Relisource Technologies Ltd.

May 2015 – Dec 2017 Dhaka, Bangladesh

mPower Social Enterprises Ltd.

Aug 2014 – Apr 2015 Dhaka, Bangladesh

Software Quality Assurance Engineer with **5+ years of experience** in manual and automated testing of web and mobile applications, contributing to large-scale systems across diverse domains.

- Supported projects delivering digital solutions for underserved populations in Bangladesh.
- Worked on web-based learning platforms for healthcare professionals and health tracking modules for developmentally disabled individuals.